Transportation Security Administration

Attachment 3: Questions to Accompany the Advanced Imaging Technology Request for Information

General Systems Description:

1. Describe the system's operating principle(s).
    a. Provide summary of enabling technology or technologies employed, including frequency range used during scanning, passenger pose position, ADA compliant, etc).
    b. Provide block diagrams/schematic illustrating major system components and functionality.
2. What is the system's source power output, and is it adjustable?
3. What are the dimensions of the system (length, width, height, footprint ($ft.^2$)?
4. Clearance requirements (maintenance, gap from ceiling, etc.)
5. What is the weight of the system?
6. Component dimensions – (standalone monitor) (if applicable)
    a. Length, width, height, quantity

Detection and False Alarm Rates:

7. Provide a list of threats (both metallic and non-metallic (explosives)) the system detects and their limits of detection.
8. Was the system developed based on machine learning principles?
9. Does the system have the capability to add additional explosives and chemicals to its library of detectable threats using a simple means such as loading a new detection algorithm?
10. Has the system undergone an independent test and evaluation of the system detection? If yes, please provide any test reports available.
11. What is the system's False Alarm Rate?
12. What are common nuisance alarms the system alarms on?
13. What divest protocol was used during algorithm development?

Processing Time/Throughput:

14. What is the Machine Processing and Analysis Time? Machine Processing and Analysis Time is defined as beginning when the scan button is initiated to when the system produces a result to the user.
15. What is the system's throughput?

Start-up:

16. How long is the machine's cold start-up procedure from power initiation to display of the login screen? Cold start-up occurs when the system has been turned off and then is turned on.

17. How long is the system warm start-up procedure from power initiation to display of the login screen? Warm start-up occurs when the system has been turned off for a period of less than thirty (30) minutes and then is turned on.

Audible and Visual Indicators:

18. Does the system provide any audio or visual indicators?
    a. Upon detection/non-detection of a threat.
    b. When the system is ready to scan again

System Error and Faults:

19. Does the system allow sample analysis when an error or fault is remedied?

Calibration/Verification Check:

20. Does the system perform a calibration/verification or prompt the user to perform a calibration/verification to ensure the system is calibrated properly?

21. If the system prompts the user to perform a calibration/verification, at what intervals does the system perform a calibration/verification or prompt the user to perform a verification?

22. How is the calibration/verification of the system performed?

Field Data Reporting and Recording:

23. Does the system include a Field Data Reporting System (FDRS) that can collect, analyze, store and output data?

24. Does the system FDRS accurately record and report sample event data? Sample event data is defined as all relevant information that can be associated with a sample analysis. At a minimum, the relevant information includes a timestamp for each scan, any change in algorithm switching, and data related to alarm locations on the body, including sensitive areas.

Error and Fault Messages:

25. When a system error or fault occurs, does the system provide an audible indicator and display a notification message on the display monitor?

26. Does the system track error and fault messages in the FDRS?

Reliability:

27. What is the Mean Time Between Critical Failures (MTBCF)?
    For determining the reliability of Transportation Security Equipment (TSE) in the operational environment, MTBCF considers the System Operating Time/Hours and the Number of Critical Failures. System Operating Time is the period a system is available to perform its required mission. Number of Critical Failures is the number of failures which prevents the TSE from performing its intended function and cannot be cleared by operator intervention alone; the fault requires a call to the maintenance service provider to restore the system to operation. Events that a Transportation Security Officer (TSO) can correct via system resets, simple removal of obstructions, etc., are not categorized as critical failures. MTBCF is calculated as follows:

$$\text{MTBCF} = \frac{\text{System Operating Time}}{\text{Number of Critical Failures}}$$

Maintainability:

28. What is the Mean Down Time (MDT)?
For determining the Maintainability of TSE for calculating Availability in the operational environment, the TSA considers the total operational time that a TSE is not in a condition to perform its required mission. This includes the time from the initial notification of the critical failure until the time the TSE is ready for acceptance and signoff by the TSA. Mean Down Time (MDT) is calculated by dividing the total operating time that a system is not in a condition to perform its required mission by the number of failures over a given period.

$$\text{MDT} = \frac{\text{Total Operating Time the System is Down}}{\text{Number of Failures Over a Given Period}}$$

Availability:

29. What is the system Availability?
Availability (Ao) is the percentage of time, over a period during operational hours, that TSE is available to perform its required mission. It reflects all potential causes of downtime, such as administrative downtime, logistics downtime, etc., and total TSE operational uptime. Ao is calculated as follows:

$$\text{Ao} = \frac{\text{Uptime}}{\text{Uptime} + \text{Downtime}} \times 100$$

Safety:

30. Does the system comply with *IEC standard 63391: 2024 Active millimeter-wave systems for security screening of humans - General requirements?*
31. Does the system have or use any hazardous materials such as chemical(s), radioactive sources, lasers, etc.? If yes, what type of protective equipment will be required for the operation and maintenance of the machine?
32. Does the system expose users or maintenance personnel to extreme temperatures?
33. Does the system expose users to ergonomic risk factors from awkward postures, stresses, and forces during system operation, calibration, and maintenance?

Quality:

34. Does the system incorporate *Institute of Electrical and Electronics Engineers (IEEE) N42.59 image quality standard?* If yes, describe how it is incorporated.
35. Does the system apply the practice of *National Electrical Manufacturers Association (NEMA) IIC 1 v05, Digital Imaging and Communications in Security (DICOS) standard* and image curation methodology? If yes, describe how it is applied.